Civics Group	A Level Index Number	Name (use BLOCK LETTERS)

H2

ST. ANDREW'S JUNIOR COLLEGE
2022 JC2 PRELIMINARY EXAMINATIONS

H2 BIOLOGY **9744/03**

Paper 3

Thursday 15th September 2022 2 hours

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and index number on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagram, graph or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.
Write your answers in the spaces provided on the question paper.

All working for numerical answers must be shown.

Conceptual error (C)	Data Quoting (D)	Expression (E)	Misreading the question (Q)

For Examiners' Use	
1	/30
2	/9
3	/11
4 or 5	/25
Total	**/75**

This document consists of **XX** printed pages.

[Turn over]

Answer **all** *questions.*

QUESTION 1

(a) SWI/SNF (SWItch/Sucrose Non-Fermentable) is a subfamily of ATP-dependent chromatin remodelling complexes found in eukaryotes. Products of SWI and SNF genes can destabilise histone-DNA interactions, resulting in sliding or ejection of nucleosomes, as shown in Fig. 1.1.

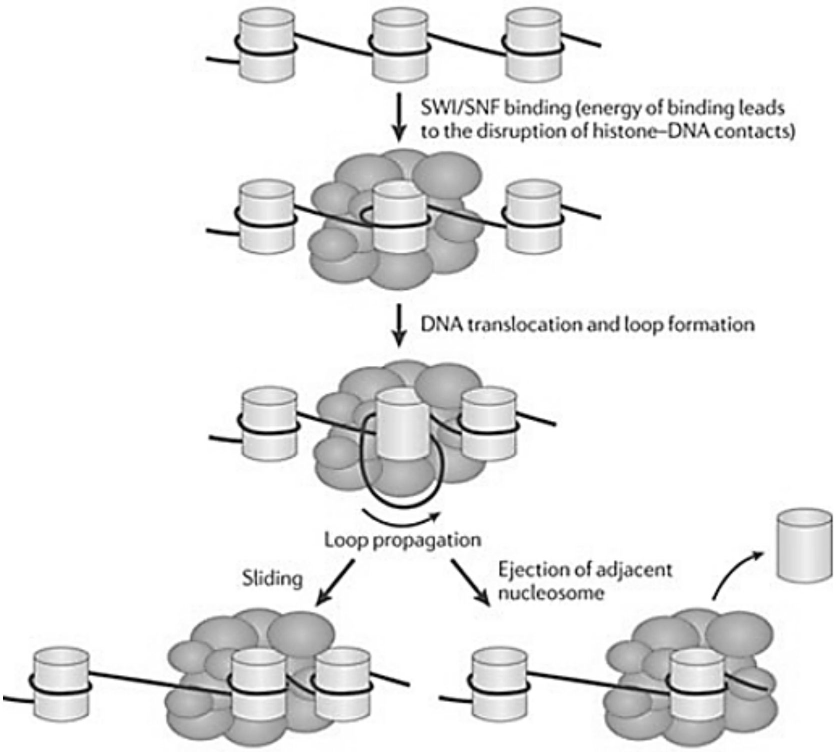


Fig. 1.1

In contrast, Polycomb Repressive Complex (PRC), with the histone methyltransferase EZH2 as its catalytic subunit, is associated with chromatin compaction.

(i) With reference to Fig. 1.1, describe how the activity of SWI/SNF complexes can lead to the active transcription of genes.

.... [2]

1. The binding of SWI/SNF complex to DNA results in **DNA translocation and loop formation** which results in sliding or ejection of nucleosomes
2. making the linker DNA accessible to transcription factors and RNA polymerase to bind and trigger transcription.

(ii) Describe two mechanisms that can produce the same chromatin remodelling effect as PRC/EZH2 complex.

....[3]

1. **DNA methylation** by DNA methyltransferases
2. Recruits histone deacetylases which **removes acetyl groups** from lysine residues in the histone tails
3. Restores ionic interaction between positively charged histone tails and negatively charged DNA, DNA becomes more compact.

(iii) Describe how the nucleosomes will eventually pack into chromosomes during prophase of mitosis.

.... [5]

1 Interactions between the linker DNA and the histone tails of the nucleosomes on either side of the linker DNA.
2 Nucleosomes and linker DNA result in the formation of a 10nm **chromatin fibre** / **"beads-on-a-string"** structure.
3 Coils/folds the chromatin into a solenoid/30 nm chromatin fibre
4 6 nucleosomes in **one turn of the helix** of the solenoid
5 Attachment of the 30-nm chromatin fibre to a central protein scaffold at multiple locations, forming a series of radial loops (looped domains), this forms a 300-nm chromatin fibre.
6 Further packing of the radial loops form the metaphase chromosome.

(b) Over 20% of human cancers carry a mutation in SWI/SNF complex subunit genes. Fig. 1.2 shows that the inactivation of SWI/SNF causes a failure to inhibit PRC at promoters and typical enhancers (TE) of genes involved in differentiation. The residual functional SWI/SNF complexes are preferentially localized to super-enhancers (SE) of genes maintaining cell renewal. The resulting imbalance between differentiation vs. self-renewal promotes tumorigenesis (tumour formation) .

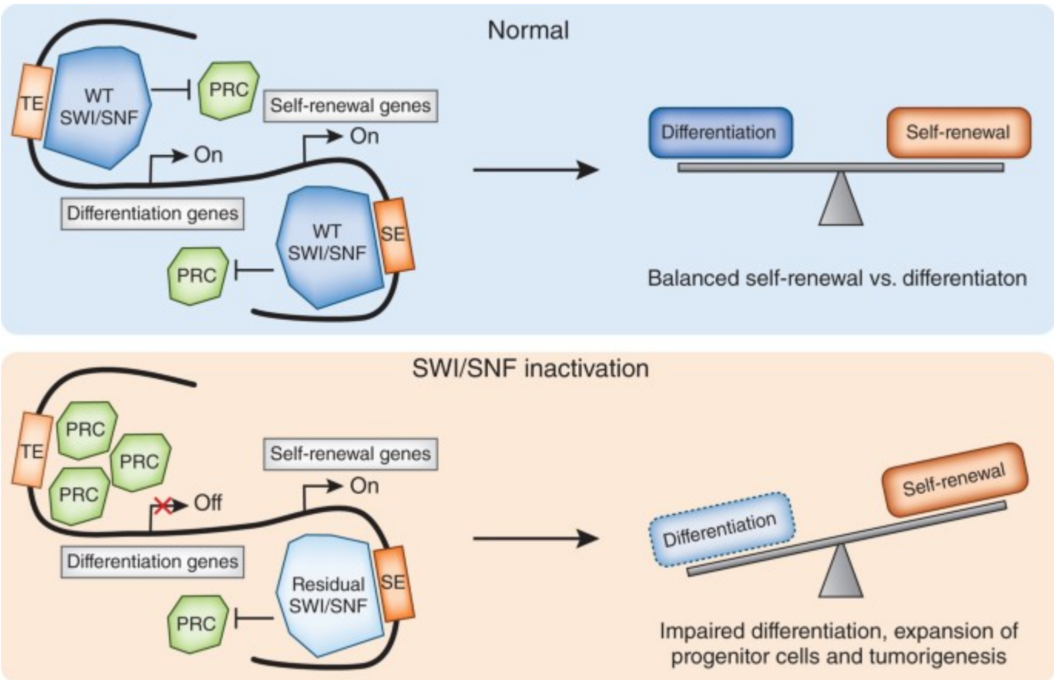


Fig. 1.2

(i) Describe the role of enhancers in the control of gene expression.

.... [3]

1 Enhancers serve as attachment site for activator proteins.
2 Increase rate of transcription of genes involved in cell differentiation.
3 Involved in time and tissue specific expression

(ii) With reference to information from **1(a)** as well as Fig. 1.2, describe the downstream effect of failing to inhibit PRC due to inactivation of SWI/SNF.

.... [3]

1 PRC leads to the **compaction of chromatin**,
2 hence general transcription factors are unable to bind to the promoter
3 and **activators are unable to bind to the typical enhancers** of genes involved differentiation.
4 Cannot stimulate/increase transcription of the genes, hence **no differentiation of the cells**.

(iii) Explain how the "imbalance between differentiation vs. self-renewal promotes tumorigenesis".

....[2]

1 The imbalance occurs when cells continue to divide **indefinitely/excessively**
2 [compulsory point]: forming a mass of **undifferentiated** /**unspecialized** cells called a tumour.

(c) Different mutations in the proto-oncogene coding for Epidermal Growth Factor Receptor (EGFR) are associated with several cancers such as lung cancer. Lung tissue samples were taken from two lung cancer patients and cultured on Petri dishes for 24 hours. The average cytoplasmic mRNA levels (Fig. 1.5A) as well as 6-hourly protein levels (Fig. 1.5B) of EGFR and β-actin were investigated. β-actin is a house-keeping gene that is expressed in almost all cell types and is used as a positive control.

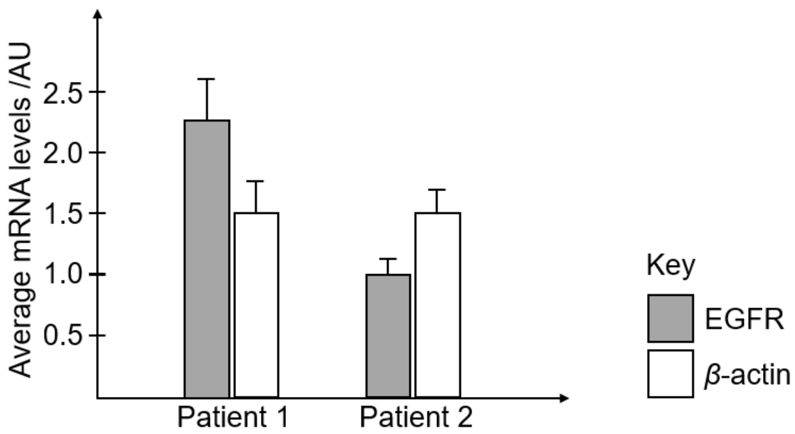


Fig. 1.5A

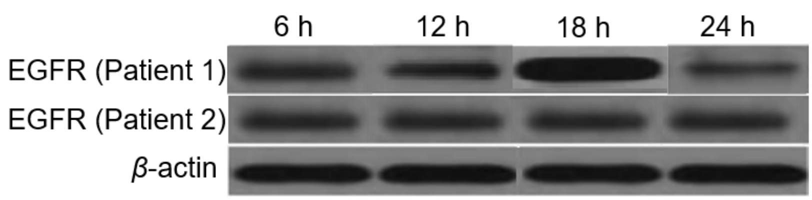


Fig. 1.5B

(i) Describe the rationale of using β-actin as a positive control.

.... [2]

1 As it is a house-keeping gene, its mRNA and protein levels are expected to **remain relatively constant** at all times;
2 It is to show that any changes in mRNA levels or protein levels in the other gene/EGFR is due to **differential gene expression and not due to any other factors**, e.g. improper lab techniques.

(ii) Describe the difference in average mRNA and protein level of EGFR between the two patients.

....[2]

1 Patient 1 has a **higher** average mRNA level of 2.25 AU compared to 1.0 AU of patient 2.
2 Protein level in patient 1 **increased from 6h to 18h** and **decreased from 18h to 24h** while protein level in patient 2 **remained relatively constant throughout the 24h**.

(iii) Suggest and explain possible mutations in the EGFR gene of patients 1 and 2 that could result in the differences described in **c(ii)**.

....[4]

Possible mutations in patient 1:

1 *[Suggest]* **Gene amplification** of EGFR resulted in **more copies of DNA templates**,
2 *[Explain]* resulting in increased transcription and higher mRNA levels.
OR
3 *[Suggest]* Mutation in the promoter of EGFR gene
4 *[Explain]* leading to higher affinity with RNA polymerase, hence increasing the rate of transcription
OR
5 *[Suggest]* **Longer poly(A) tail** is added to the mRNA of patient 1,
6 *[Explain]* resulting in **longer half-life**, and hence higher cytoplasmic mRNA levels.

Possible mutation in patient 2:

7 *[Suggest]* **Substitution** mutation in the **coding sequence**
8 *[Explain]* resulted in oncoprotein that is **resistant to degradation** as seen from no decrease in protein levels
OR
9 *[Suggest]* **Substitution** mutation in **coding sequence**
10 *[Explain]* resulted in (normal expression level of the receptor but the) **hyperactive** receptor which can trigger cell signaling **despite absence of ligand**

(iv) Contrast between the function of a proto-oncogene such as EGFR and a tumour suppressor gene such as p53.

....[2]

1 EGFR codes for a receptor involved in **cell signalling**, stimulating **normal cell division**
2 while p53 codes for a transcription factor that **triggers the expression of other genes** whose **products arrest cell cycle, repairs DNA and triggers apoptosis**.

(v) The tissue samples taken from the two lung cancer patients showed mutations in several other genes besides proto-oncogenes and tumour suppressor genes.

Explain how these mutations contributed to the development of lung cancer.

....[2]

Some mutations cause

1 activation of telomerase gene, which codes for telomerase enzyme which restores the telomeres, allowing cancer cells to divide indefinitely
2 the loss of ability to differentiate
3 loss of anchorage-dependent inhibition / loss of cell adhesion
4 loss of contact-dependent / density-dependent inhibition
5 angiogenesis, the recruited blood vessels provide cancer cells with oxygen and nutrients for growth and to remove any waste products.
6 metastasis / cancer cells detach from the parent mass, invade surrounding tissues and spread via the circulatory system to other tissues to form secondary tumors

Reject mutations in telomerase gene

[Any 2]

[Total: 30]

QUESTION 2

Measles is an infectious disease for which vaccines have been developed. The commonly used vaccine consists of an attenuated (weakened) form of the virus. The measles vaccine is normally given to children when they are about one year old, followed by a booster dose when they are about four years old.

Fig. 2.1 shows the number of reported cases of measles and the percentage of the population vaccinated worldwide between 1980 and 2002.

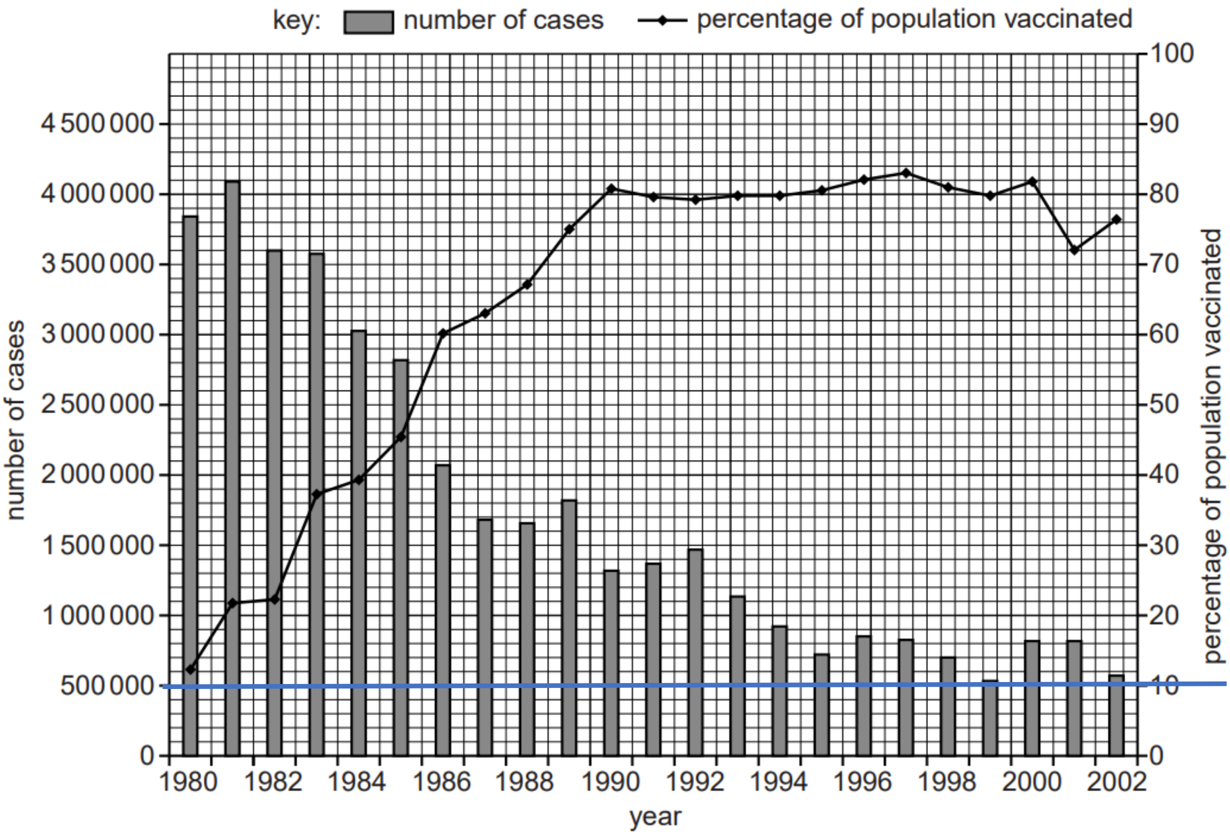


Fig. 2.1

(a) Describe the trends shown in Fig. 2.1:

between 1980 and 1990

....

between 1990 and 2002.

....[4]

1980 – 1990

- percentage of population vaccinated increased from 12% to 81% ;
- number of cases decreased (steeply) from 3850000 to Accept :130000-1320000 ;

1990 – 2002

- percentage vaccinated, levels off/remains relatively constant between 79-81% from 1990-2000 and decreasing to 72% in 2001 before increasing to 76% in 2002 /relatively constant;

- number of cases decreases (less steeply than earlier) to 700,000 in 1994 and fluctuates between 500,000 to 800,000 from 1995 to 2002 ;

(b) The vaccination of a high enough percentage of the population can break the disease transmission cycle, giving 'herd immunity'.

The measles virus spreads easily by droplet transmission. One estimate of its reproduction number, R_o, is 15, meaning that each person infected with measles is expected to infect 15 other people.

If the percentage of people who are immune to a disease exceeds the herd immunity threshold, the disease can no longer persist in the population. Assuming 100% efficacy of the vaccine, the herd immunity threshold is calculated as:

$$100 \times (1 - 1/R_o)$$

(i) Calculate the herd immunity threshold for measles, if R_o = 15.

Herd immunity threshold = ……………………….% [1]

ANS:
Herd immunity threshold = 93.3%

(ii) Use Fig. 2.1 and your answer to **(b)(i)** to explain why the number of measles cases remain above 500,000.
………………………………………………………………………………………….[2]

1. According to Fig 2.1, **the highest percentage of the population vaccinated was in 1997 where 83 % of the population was vaccinated**. The percentage of vaccination also fell to 72% in 2001.
2. This was **lower than the herd immunity threshold of 93.3%** and therefore the measles virus is still able to spread in the population causing the number of measles cases to remain above 500,000

(c) The measles virus has a unique protein on its surface called MV-H which can bind to a protein called CD-46 on the surface of human cells. This allows the measles virus to infect these cells.

Suggest how the two proteins, MV-H and CD-46, can bind to each other.

…………………………………………………………………………………………[2]

1 CD–46 is a receptor on the cell surface which has binding site which is **complementary in shape/conformation** of MV-H;

2 ref. to interaction of **R-groups** /amino acid side chains in the binding site with the R-groups of MV-H ;

3 ref. formation of hydrogen bonds / ionic bonds to bind MV-H and CD-46 together.

REJECT disulfide/ peptide

[Total: 9]

QUESTION 3

Fig. 3.1 shows the percentage cover of live corals and the density of herbivorous fish (plant-feeding) on a coral reef over a number of years. Due to unusually warm water, many of the corals living on the reef died in 1998.

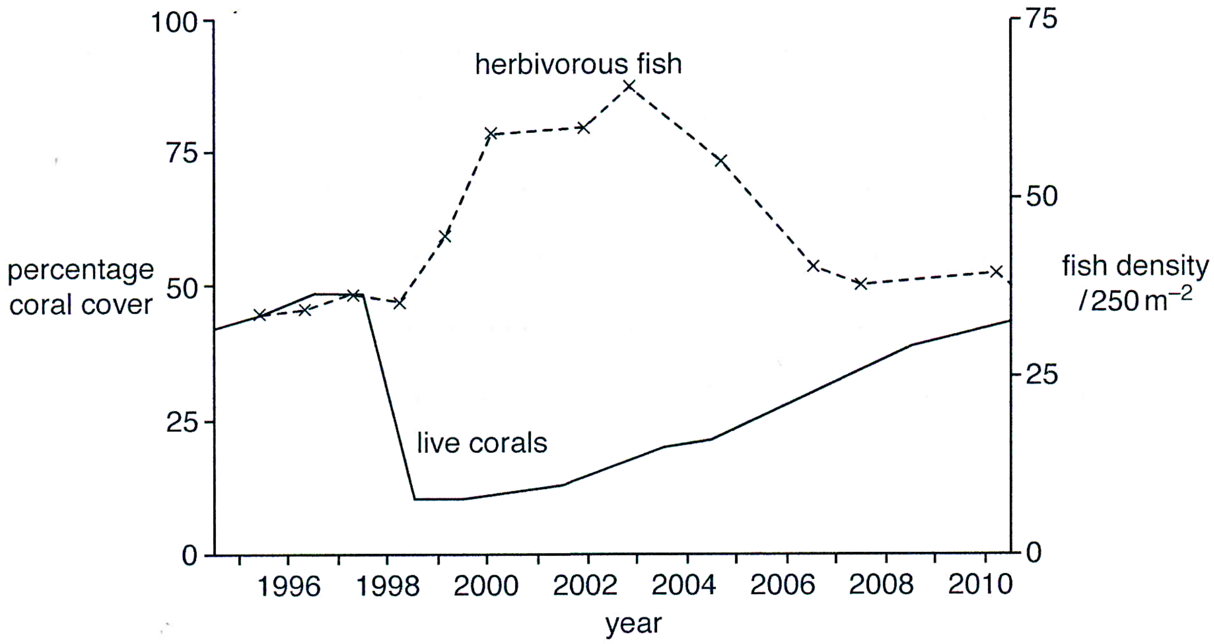


Fig. 3.1

(a) Describe the change in percentage cover of live corals shown in Fig. 3.1.
....[3]

1. Slight **increase** in percentage coral cover from around **40% to 50%** observed from **mid 1994 till mid 1997**,
2. Followed by a sharp **decrease/dip** in percentage coral cover (from 50%) **to 10% till mid 1998,**
3. Percentage coral cover start to **increase** steadily (from 10%) **to around 40% after mid 1998** to year 2010;

(b) Suggest possible impacts due to the change in percentage cover of the live corals.
....[3]

1. Possible shift in food sources to other plant sources (i.e. sea grass and kelps) resulting in higher competition with other species feeding on these food sources.
2. Population size of species dependent on corals as habitats will decrease as they may die / migrate to other coral reefs /OWTTE;
3. Loss of biodiversity;
4. Direct correlation in change of coastal protection in tropics region;

5 Change in global food supply distribution of fishes that depends on corals;
6 Shift in ecotourism / tourism;

(c) State a possible reason why the density of herbivorous fish increases after the death of many of the corals in 1998.

....[1]

1 ref. death of corals due to the unusually warm water leading to expulsion of more zooxanthellae/algae for herbivorous fish to feed on

2 possible change increase in growth of sea grass and kelps due to increase in temperature

The impact of climate change is a major threat not only for corals but also for life on earth a whole. Fig. 3.2a shows the loss of ice mass between 1992 and 2018. Fig. 3.2b shows the median extent of sea ice concentration in 1992 as compared to 2018. Fig. 3.3 the number of polar bears living in this biome.

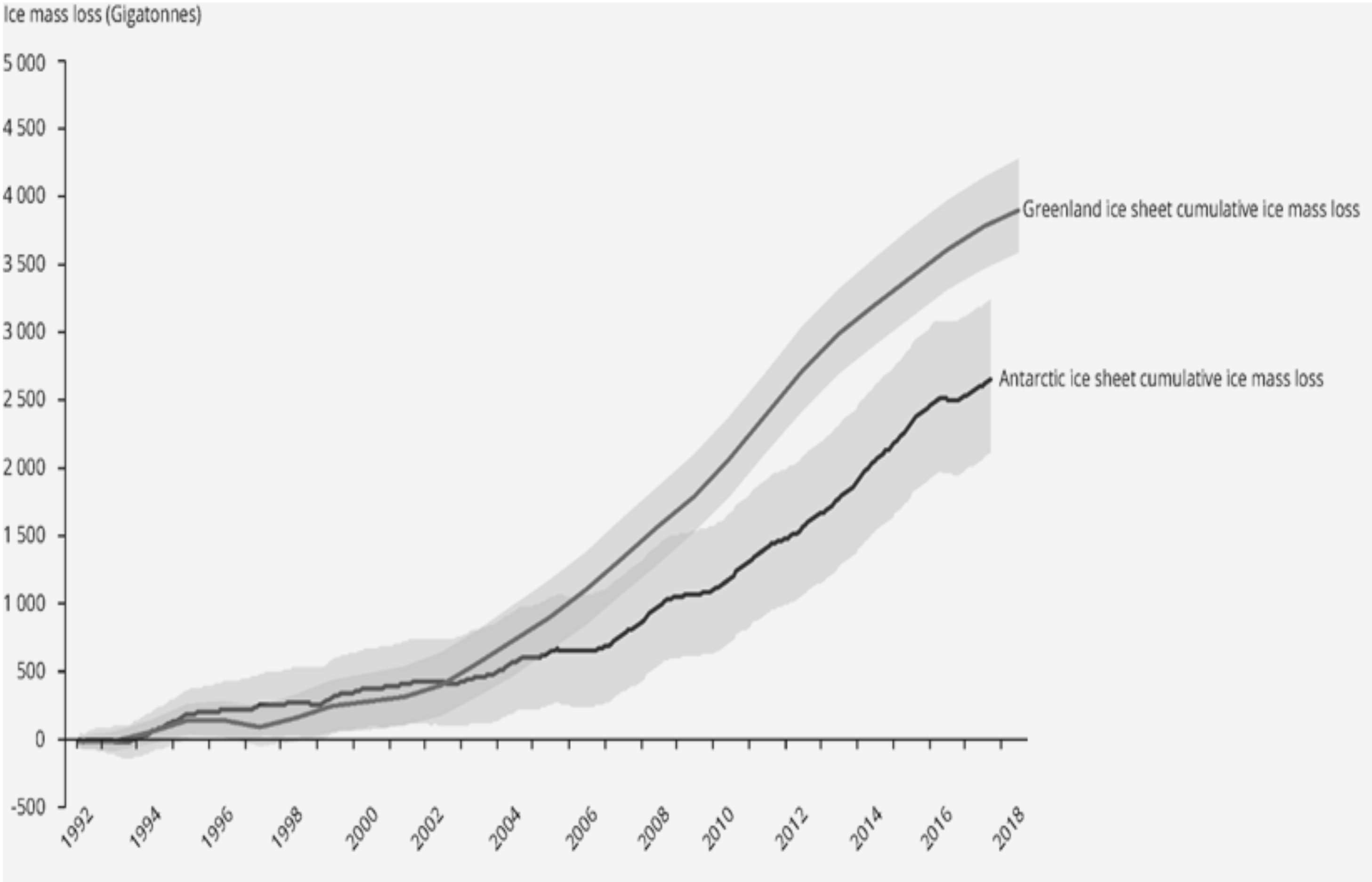
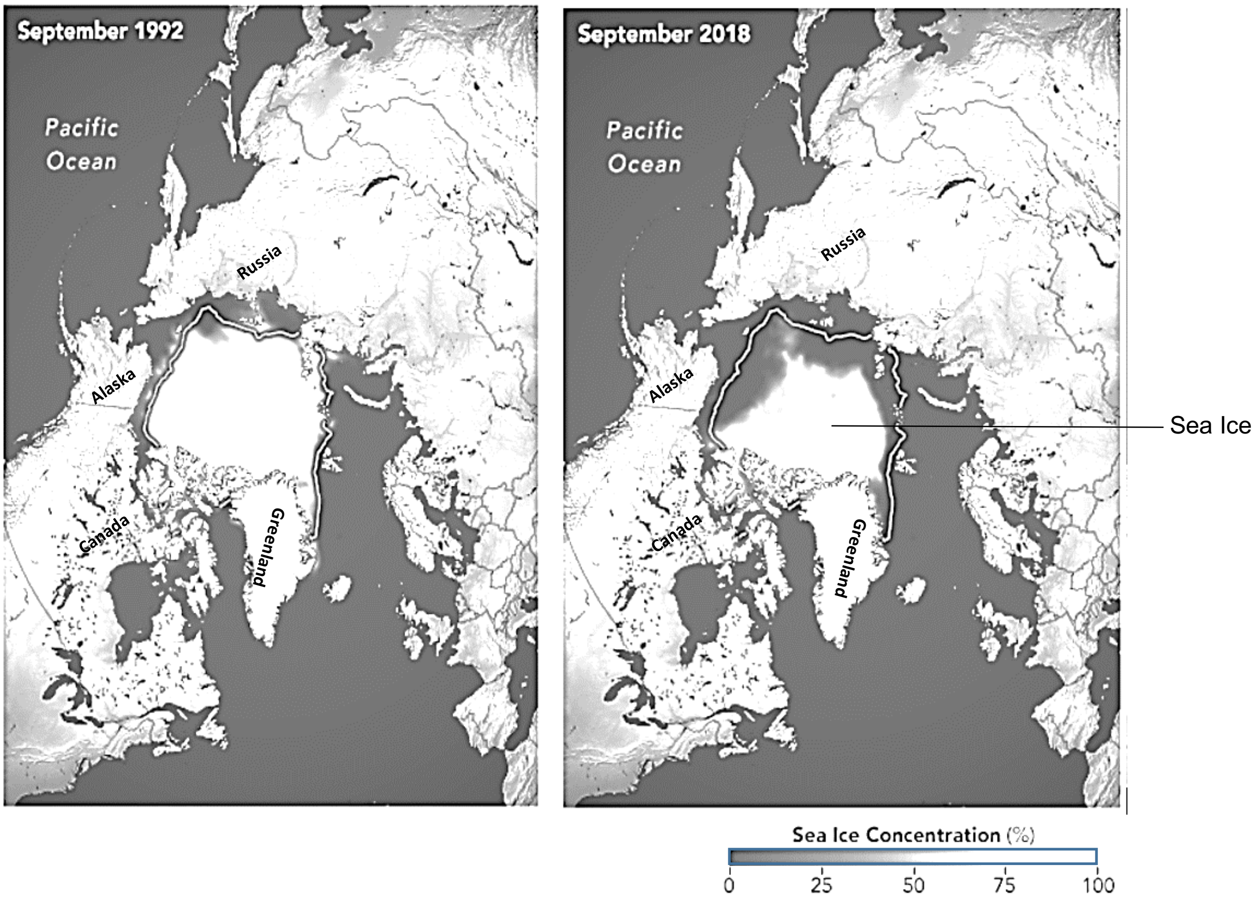


Fig. 3.2a


September 1992
Pacific Ocean
Russia
Alaska
Canada
Greenland
September 2018
Pacific Ocean
Russia
Alaska
Canada
Greenland
Sea Ice
Sea Ice Concentration (%)
0
25
50
75
100

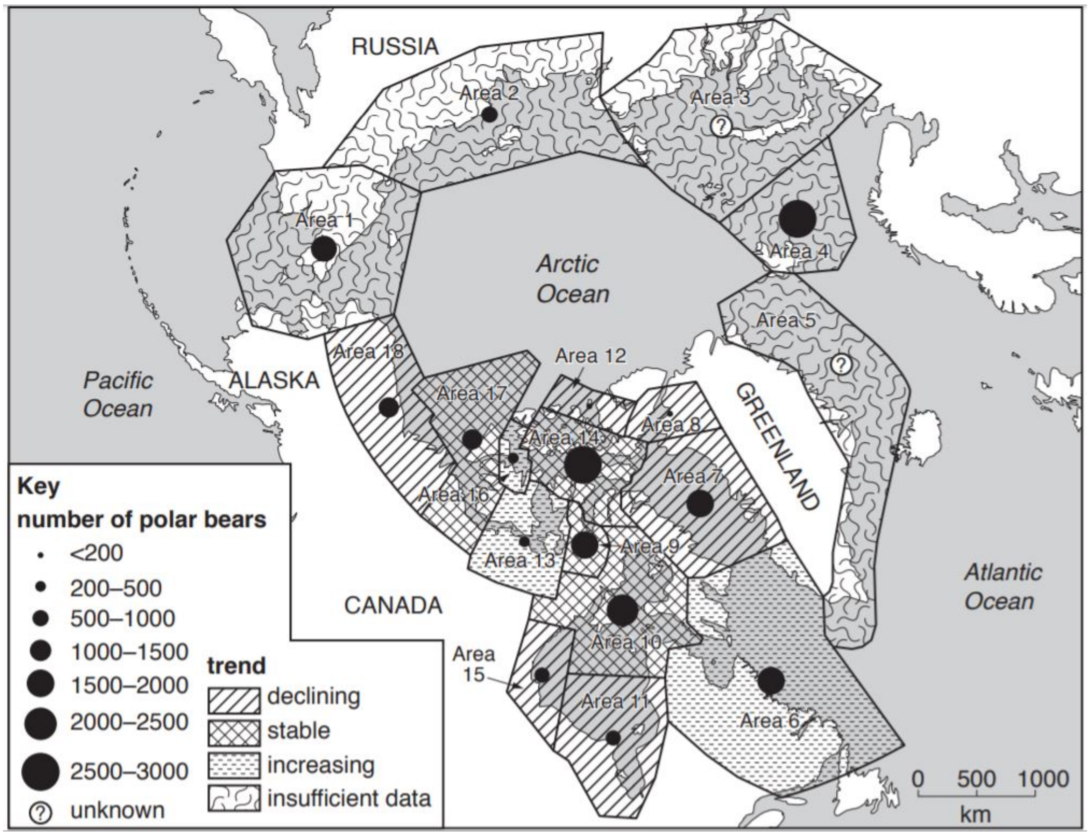

RUSSIA
Area 2
Area 3
Area 1
Area 4
Arctic Ocean
Area 5
Area 18
Area 12
Pacific Ocean
ALASKA
Area 17
Area 14
Area 8
GREENLAND
Area 16
Area 7
Area 13
Area 9
CANADA
Atlantic Ocean
Area 10
Area 15
Area 11
Area 6
Key
number of polar bears
<200
200–500
500–1000
1000–1500
1500–2000
2000–2500
2500–3000
unknown
trend
declining
stable
increasing
insufficient data
0 500 1000
km

Fig. 3.3

(d) With reference to Figs. 3.2a, 3.2b and 3.3, describe and explain the evidence which suggest that polar bears are at risk of extinction as a result of climate change.

....[4]

1 Climate change results in global warming / rising temperatures, shrinking of sea ice floating in the Arctic ocean and ice sheets on the island of Greenland as seen from Fig.3.2b

2 As seen from Fig. 3.2a, the increase in loss of ice mass from 0 to 3800 gigatonnes from 1992-2018 for Greenland

3 This contributes to habitat loss / fewer hunting grounds for polar bear

any two from:

4 which is evident from the very low populations of polar bears in some areas, e.g. less than 200 in Area 12 and Area 8 (Lancaster Sound / Kane Basin)

5 declining populations in 6 / 50% / of areas e.g. Area 7, 8, 11, 12, 15, 18

6 a population range of less than 2000 in areas where the populations are declining e.g. Area 7, 8, 11, 12, 15, 18

7 there are only a few, areas e.g. Area 6, 13, where the population is currently increasing

[Total: 11]

Essay

Answer **one** question only in this section.

Write your answers on the lined paper provided at the end of this question paper.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections (a), (b) etc., as indicated in the question.

4 **(a)** Microorganisms such as *Escherichia coli* (*E. coli*) colonise the intestine and obtain nutrients from their surroundings. [15]

Describe how *E. coli* responds to the presence of lactose and absence of glucose in the intestine and explain how a mutation in the regulatory sequences of the *lac* operon may affect how *E. coli* respond to changes in lactose supply.

(b) The green fluorescent protein (GFP) gene was isolated from the jellyfish *Aequorea Victoria* and inserted into DNA of a species of mice, *Mus musculus*. The gene was expressed throughout the genetically modified mice resulting in the mice exhibiting bright green fluorescence when exposed to light in the blue to ultraviolet range. [10]

With the use of the various species concepts, discuss whether the genetically modified mice can be considered as the same or different species as the unmodified mice.

5 **(a)** 'Eukaryotic life is impossible without membranes.' [15]

Discuss how regulation of membrane fluidity is achieved in living organisms and justify why the statement above may be true using specific examples.

(b) With reference to named examples, describe the role of water in reactions involving biomolecules. [10]

End of Paper

4(a) Microorganisms such as *Escherichia coli* (*E. coli*) colonise the intestine and obtain nutrients from their surroundings.

Describe how *E. coli* respond to the presence of lactose and absence of glucose in the intestine and explain how a mutation in the regulatory sequences of the *lac* operon may affect how *E. coli* respond to changes in lactose supply. [15]

10 green 5 yellow

[Describe how *E coli* responds to changes] [Max 8]
[Presence of lactose]

1 Lactose **isomerizes** to form allolactose;
2 ref. allolactose as an inducer;
3 allolactose binds to the (allosteric site of) the lac repressor, and **alters its 3D conformation** (at the DNA-binding site);
4 lac repressor become **inactive** and is no longer able to bind to the operator (lacO);
5 RNA polymerase can **access and transcribe** the **structural genes**;
6 ref. lacZ , lacY and lacA genes being transcribed and translated into β-galactosidase, lac permease and lactose transacetylase enzymes respectively;

[Absence of glucose]

7 **cyclic adenosine monophosphate / cAMP concentration increases** and binds to (allosteric site of) catabolite activator protein (CAP) site;
8 alters its **3D conformation** of the (DNA-binding site of) CAP;
9 CAP is **active** and binds to the CAP-binding site;
10 ref. facilitates more **efficient positioning of RNA polymerase** at the promoter;
11 **High rate of transcription** of lac operon;

[Effect of mutation of regulatory sequences] [Max 6]
Max 3
12 Mutation of the **promoter** results in a **change in the sequence/structure/shape** of the promoter;

Mutation of lac promoter (increases binding affinity)
13 ref. **RNA polymerase** able to **bind** to the (mutated) promoter with **greater affinity**; [Reject: permanently/ irreversibly]
14 … hence **increase the rate of the transcription** of structural genes **in the presence of lactose**;
OR
Mutation of lac promoter (decrease binding affinity)
15 ref. **3D shape/conformation** of the promoter is **no longer complementary** to the (active site) of **RNA polymerase**;
/ RNA polymerase **no longer able to bind** to the (mutated) promoter;
16 ref. **No transcription** of structural genes **even in the presence of lactose**;

Max 3
Mutation of operator
17 Mutation of the **operator** results in a **change in the sequence/structure/shape** of the operator;

18 ref. repressor binds to the operator **permanently/irreversibly;**
19 ref. **No transcription** of structural genes **even in the presence of lactose**;
OR

20 ref. **3D shape/conformation** of the operator is **no longer complementary** to the of the **repressor**;
/ (active) repressor can **no longer bind** to the operator;

21 ref. RNA polymerase is now able to **access / bind** to promoter, **allowing transcription** of structural genes **even in the absence of lactose**.

QWC [1]
Accurate communication of the **response to lactose and absence of glucose** in *E.coli*, and to include at least **one mutation in both promoter and operator**

4(b) The green fluorescent protein (GFP) gene isolated from the jellyfish *Aequorea Victoria* and inserted into DNA of a species of mice, *Mus musculus*. The gene was expressed throughout the genetically modified mice resulting in the mice exhibiting bright green fluorescence when exposed to light in the blue to ultraviolet range.

With reference to the various species concepts, discuss whether the genetically modified mice can be considered as the same or different species with the unmodified mice. [10]

Biological species concept

1 [*Definition*] Closely related organisms which are **capable of interbreeding in nature** to produce viable, fertile offspring and are **reproductively isolated from other species**;

2 [*Support for same species argument*]
ref. Coat colour of GM mice not affected as green fluorescent protein genes only affect colour of the mice under UV light, thus **traits which affect their ability to interbreed** with non-GM mice not affected**;**

3 [*Does not support same species argument*]
ref. sexual selection, i.e. non-GM mice that do not fluoresce under UV light prefer to breed with other non-GM mice;
/ preferential breeding of specific mice with colour, hence GM mice do not interbreed with non-GM mice;

Genetic species concept

4 [*Definition*] Group of **genetically compatible (i.e. similar DNA) interbreeding natural populations** that are **genetically isolated** from other such groups.

5 [*Support for same species argument*]
ref. GM and non-GM mice being **genetically similar** / low genetic variation / do not have sufficient genetic differences/distance / difference only in 1 gene (i.e. the GFP gene from jellyfish);

6 [*Does not support same species argument*]
ref. sufficient genetic differences (an entire additional gene not present in original species);

Phylogenetic species concept

7 [*Definition*] The smallest group of organisms that shares a **most recent common ancestor** forming one branch on the tree of life.
/ Defines a species based on their genetic history and **evolutionary relationships**, with reference to their homologous structures, and nucleotide and protein sequences;

8 [*Support for same species argument*]
ref. both GM and non-GM mice form the smallest group that shares a same / recent common ancestor;
/ ref. comparison of homologous structures or high homology of nucleotide/protein sequences;

Morphological species concept

9 [*Definition*] A group of organisms sharing **similar body shape, size** and other **structural/ physical (anatomical) features**;

10 [*Support for same species argument*]
GM and non-GM mice **look** largely **similar** to each other based on **physical characteristics**;

11 [*Does not support same species argument*]
ref. coat colour of GM and non-GM mice is different under UV light;
/ GM mice look very different from non—GM mice due to the green florescence;

Ecological species concept

12 [*Definition*] Organisms sharing the <u>same ecological niche</u> (i.e its unique and particular role in an ecosystem);

13 [*Support for same species argument*]
GM and non-GM mice still occupy the same ecological niche, ref. predator-prey relationship / food source / position in food chain etc (give at least one example);

Accept argument if support opposing viewpoints in different species concept/same species concept.

QWC [1]
Logical flow of information + **at least 3 species concepts** to support argument

5(a) 'Eukaryotic life is impossible without membranes.' Discuss how regulation of membrane fluidity is achieved in living organisms and justify why the statement above may be true using specific examples. [15]

How membrane fluidity is regulated [max 5]

1 Dependent on the **proportion of phospholipids** with unsaturated hydrocarbon/fatty acid tails and cholesterol;

[At low temperatures]

2 ref. Unsaturated hydrocarbon/fatty acid tails have kinks that prevent phospholipids from **close packing**, enhancing membrane fluidity; (Accept reverse argument)

3 Cholesterol prevents **close packing of phospholipids** in the membrane, preventing membrane from freezing;

[At high temperatures]

4 Membrane prevented from being overly **fluid** / **integrity of the membrane is maintained** as cholesterol **restricts phospholipid movement**;

5 ref. hydrophobic interactions between cholesterol and fatty acids tails, stabilize lipid bilayer;
/ ref. hydrophobic interactions between fatty acids stabilise the membrane;

6 ref. length of fatty acid tails also helps maintain membrane fluidity, since longer tails will result in **more hydrophobic interactions** between the tails and hence, decrease membrane fluidity (Accept reverse argument);

Diverse roles of membranes in named processes [max 10]

Note: Students should be able to phrase their answer as to how eukaryotic life is impossible without membranes.

e.g. Without membranes, compartmentalisation within a cell cannot take place. Hence, it will be impossible to concentrate enzymes and substrates into a localised area to speed up the rate of biochemical pathways.

e.g. It will also be impossible to establish a concentration gradient between two compartments, such as in mitochondria and chloroplasts, making it impossible for chemiosmosis and ATP synthesis to occur.

e.g. It will be impossible to separate molecules involved in different biochemical pathways from each other, hence they either cannot occur simultaneously.

Context	**Roles**
Selectively permeable / S	**7** Regulates movement of substances into and out of cell; **[Max 1]** **8** Non-polar/uncharged molecules are able to dissolve and diffuse through; **9** Ions/charged and/or most polar molecules are repelled; (Generic principle cannot be replaced by H^+)
Compart-mentalisation / C	1. Formation of unique environment for highly specialised activities in the cells; 2. e.g. lysosomes maintain an acidic environment that favours its enzymes to work;

	3. localization of enzymes for reactions to take place so that they are not suppose to catalyse reactions that they are not supposed to; 4. e.g.: enzymes in Calvin Cycle in stroma; 5. accumulation of ions to generate a concentration gradient; 6. e.g. proton gradient established across a named memberane/H^+ in intermembranal space in mitochondria/within thylakoid space/ in chloroplasts establishes a proton gradient for *chemiosmosis** and formation of ATP; (no need to talk abt atp synthase if mention abt chemiosmosis) 7. Storage of food source; 8. e.g. starch in plant cells are stored in amyloplasts; R: starch grain, make sure it is membrane bound
Localisation of proteins of a related function / L	9. Allows functionally-related proteins to be grouped together to enhance sequential biochemical processes; 10. e.g. enzymes and proteins are grouped into photosystems II and I on thylakoid membrane of chloroplast so that electrons from photosystem II are shuttled to photosystem I during photophosphorylation;
Increase surface area / I	11. Highly folded increases surface area to hold more; 12. e.g.: electron transport chains/ ATP synthase in cristae/inner mitochondrial membrane of mitochondria;
Cell-cell recognition and adhesion / R	13. Glycoproteins / glycolipids are involved in cell-cell recognition; 14. (any 1 e.g.) eg: T cell receptor on membranes of naïve T cells + how it helps to recognize peptide: MC on antigen presenting cells;
Signal transduction / T	15. Some transmembrane proteins serve as cell surface receptors to transfer information from environment into cell when specific molecules (ligands) bind to them; 16. E.g. glucagon (hormone) bind to glucagon receptor in membrane which triggers a cascade of chemical reactions leading to hydrolysis of glycogen to glucose; E.g. insulin

QWC: explanation of one mechanism of regulation of membrane fluidity + ref to 3 different context

5(b) With reference to named examples, describe the role of water in reactions involving biomolecules. [10]

1 [role] ref. Condensation reactions to form covalent bonds involving loss of water molecule;
2 [role] ref. Hydrolysis reactions to break bonds using water molecule;

Accept these condensation/hydrolysis of these example
3 α(1,4) and α(1,6) glycosidic bonds between α glucose molecules in amylose, amylopectin;
4 β(1,6) glycosidic bonds between β-glucose residues in cellulose;
5 peptide bonds between amino acids in proteins;
6 ester bond between glycerol and fatty acid in lipids;
7 phosphodiester bonds between deoxyribonucleotides / ribonucleotides in DNA / RNA;
AVP Accept labelled drawing/diagram

8 [role] ref. metabolic water as a solvent for reactions involving biomolecules to take place;
9 i.e. aqueous medium in cytoplasm for glycolysis to occur / stroma/matrix for enzymatic reactions in photophosphorylation/oxidative phosphorylation;
10 i.e. aqueous medium in RBCs for folding of haemoglobin / packing into RBCs;
AVP transport of sucrose

11 [role + example] photolysis of water for non-cyclic photophosphorylation which generates ATP and NADPH for formation of carbohydrates in Calvin cycle;

QWC: Logical flow of information + At least 3 different named examples describe 2 different roles of water